

[illegible]

```
[10]: import spacy
      nlp = spacy.load("en_core_web_sm")
      doc = nlp("Hello, I am learning spaCy.")
      print([token.text for token in doc])

      ['Hello', ',', 'I', 'am', 'learning', 'spaCy', '.']
```

```
[12]: pip install gensim
```

```
Defaulting to user installation because normal site-packages is not writeable
Collecting gensim
  Downloading gensim-4.4.0-cp313-cp313-win_amd64.whl.metadata (8.6 kB)
Requirement already satisfied: numpy>=1.18.5 in c:\programdata\anaconda3\lib\site-packages (from gensim) (2.1.3)
Requirement already satisfied: scipy>=1.7.0 in c:\programdata\anaconda3\lib\site-packages (from gensim) (1.15.3)
Requirement already satisfied: smart_open>=1.8.1 in c:\users\vishw\appdata\roaming\python\python313\site-packages (from gensim) (8.0.1)
Requirement already satisfied: wrapt in c:\programdata\anaconda3\lib\site-packages (from smart_open>=1.8.1->gensim) (1.17.0)
Downloading gensim-4.4.0-cp313-cp313-win_amd64.whl (24.4 MB)
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```

```
[1]: from gensim import corpora

# 1. Sample tokenized documents
documents = [
    ["natural", "language", "processing"],
    ["language", "models", "word", "embeddings"]
]

# 2. Build dictionary
dictionary = corpora.Dictionary(documents)

# 3. Convert documents to Bag-of-Words vectors (the corpus)
corpus = [dictionary.doc2bow(doc) for doc in documents]

print(corpus)

[[{(0, 1), (1, 1), (2, 1)}, {(0, 1), (3, 1), (4, 1), (5, 1)}]
```

```
[2]: corpus = [[(0, 1), (1, 1), (2, 1)], [(0, 1), (3, 1), (4, 1), (5, 1)]]

# Print actual words for the first document:
doc_0 = corpus[0]
decoded_doc = [(dictionary[word_id], count) for word_id, count in doc_0]

print(decoded_doc)

[('language', 1), ('natural', 1), ('processing', 1)]
```

```
[3]: pip install torch
```

```
Defaulting to user installation because normal site-packages is not writeableNote: you may need to restart the kernel to use updated packages.

Collecting torch
  Downloading torch-2.13.0-cp313-cp313-win_amd64.whl.metadata (39 kB)
Requirement already satisfied: filelock in c:\programdata\anaconda3\lib\site-packages (from torch) (3.17.0)
Requirement already satisfied: typing-extensions>=4.10.0 in c:\programdata\anaconda3\lib\site-packages (from torch) (4.12.2)
Collecting setuptools>=77.0.3 (from torch)
  Downloading setuptools-83.0.0-py3-none-any.whl.metadata (6.6 kB)
Requirement already satisfied: sympy>=1.13.3 in c:\programdata\anaconda3\lib\site-packages (from torch) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in c:\programdata\anaconda3\lib\site-packages (from torch) (3.4.2)
Requirement already satisfied: jinja2 in c:\programdata\anaconda3\lib\site-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in c:\programdata\anaconda3\lib\site-packages (from torch) (2025.3.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\programdata\anaconda3\lib\site-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in c:\programdata\anaconda3\lib\site-packages (from jinja2->torch) (3.0.2)
```

```
[4]: pip install torch --index-url https://download.pytorch.org/whl/cu121
```

```
Defaulting to user installation because normal site-packages is not writeable
Looking in indexes: https://download.pytorch.org/whl/cu121
Requirement already satisfied: torch in c:\users\vishw\appdata\roaming\python\python313\site-packages (2.13.0)
Requirement already satisfied: filelock in c:\programdata\anaconda3\lib\site-packages (from torch) (3.17.0)
Requirement already satisfied: typing-extensions>=4.10.0 in c:\programdata\anaconda3\lib\site-packages (from torch) (4.12.2)
Requirement already satisfied: setuptools>=77.0.3 in c:\users\vishw\appdata\roaming\python\python313\site-packages (from torch) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in c:\programdata\anaconda3\lib\site-packages (from torch) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in c:\programdata\anaconda3\lib\site-packages (from torch) (3.4.2)
Requirement already satisfied: Jinja2 in c:\programdata\anaconda3\lib\site-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in c:\programdata\anaconda3\lib\site-packages (from torch) (2025.3.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\programdata\anaconda3\lib\site-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in c:\programdata\anaconda3\lib\site-packages (from Jinja2->torch) (3.0.2)
Note: you may need to restart the kernel to use updated packages.
```

```
[5]: pip install transformers huggingface_hub datasets
```

```
Defaulting to user installation because normal site-packages is not writeableNote: you may need to restart the kernel to use updated packages.
```

```
Collecting transformers
  Downloading transformers-5.14.1-py3-none-any.whl.metadata (32 kB)
Collecting huggingface_hub
  Downloading huggingface_hub-1.26.0-py3-none-any.whl.metadata (16 kB)
Collecting datasets
  Downloading datasets-5.0.1-pv3-none-any.whl.metadata (23 kB)
```

```
[6]: from transformers import pipeline
```

```
# Load a quick sentiment analysis pipeline
classifier = pipeline("sentiment-analysis")
result = classifier("Installing Hugging Face is so easy!")

print(result)
```

```
[transformers] No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.
config.json: 0%|          | 0.00/629 [00:00<?, ?B/s]
```

C:\Users\vishw\AppData\Roaming\Python\Python313\site-packages\huggingface_hub\file_download.py:141: UserWarning: `huggingface\_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\vishw\cache\huggingface\hub\models--distilbert--distilbert-base-uncased-finetuned-sst-2-english. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF\_HUB\_DISABLE\_SYMLINKS\_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.

To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: <https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development>

```
warnings.warn(message)
model.safetensors: reconstructing file: 0%|          | 0.00B / 268MB
model.safetensors: downloading bytes:          | 0.00B
Loading weights: 0%|          | 0/104 [00:00<?, ?it/s]
tokenizer_config.json: 0%|          | 0.00/48.0 [00:00<?, ?B/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
vocab.txt: 0%|          | 0.00/232k [00:00<?, ?B/s]
```

ert--distilbert-base-uncased-finetuned-sst-2-english. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF\_HUB\_DISABLE\_SYMLINKS\_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.

To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: <https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development>

```
warnings.warn(message)
model.safetensors: reconstructing file: 0%|          | 0.00B / 268MB
model.safetensors: downloading bytes:          | 0.00B
Loading weights: 0%|          | 0/104 [00:00<?, ?it/s]
tokenizer_config.json: 0%|          | 0.00/48.0 [00:00<?, ?B/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
vocab.txt: 0%|          | 0.00/232k [00:00<?, ?B/s]
[{'label': 'POSITIVE', 'score': 0.9889538288116455}]
```

```
[8]: from transformers import pipeline
classifier = pipeline("sentiment-analysis")
result = classifier(" I love Natural Language Processing!")
print(result)
```

```
[transformers] No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.
```

```
Loading weights: 0%|          | 0/104 [00:00<?, ?it/s]
[{'label': 'POSITIVE', 'score': 0.9998558759689331}]
```

```
[ ]:
```